

A Project Proposal On
Library Management System using Sockets

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Submitted To:

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Subject:

Network Programming (NP)

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PLATFORM AND LANGUAGES:

Unix Sockets in C .

OVERVIEW:

This project implements a basic client-server library management system using TCP sockets. Multiple clients connect to a server to perform simple library operations.

OBJECTIVES:

- Learn socket programming.
- Implement client-server communication.
- Handle multiple clients using threads.
- Manage book records.
- Apply basic authentication.

FEATURES:

1. Login with username/password.
2. View books.
3. Search books.
4. Borrow book.
5. Return book.
6. Exit.

IMPLEMENTATION:

Server: Unix TCP socket, pthreads for concurrency, file-based storage.

Client: Console application sending requests and receiving responses.

SOCKET FUNCTIONS:

- socket(), bind(), listen(), accept(), connect(), send(), recv(), close().

I/O MODEL:

Blocking sockets.

CONCURRENCY:

One thread per client using pthreads.

BASIC SECURITY:

Username/password authentication and input validation.

DEVELOPMENT STEPS:

1. Create server.
2. Create client.
3. Add authentication.
4. Implement library operations.
5. Test with multiple clients.

EXPECTED OUTCOMES:

A working socket-based library server demonstrating networking, concurrency, and basic security.